Sabaa at NADI 2026 Subtask 1.1: Matching Annotation Conventions for Dialectal Arabic ASR

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Abstract

This paper describes our submission to NADI 2026 Subtask 1.1, speech recognition for eight Arabic dialects under noisy conditions. Our system adapts `whisper-large-v3-turbo` with one low-rank adapter per dialect, decodes with tuned beam search, and applies a normalization step that matches the spelling conventions of the reference transcriptions rather than applying generic orthographic rules. We train only on the shared-task data, which places our results well below those of systems built with large external corpora, and we report our numbers with that constraint in mind. Country-average WER falls from 0.804 for the zero-shot backbone to 0.6053 on the test set. We also report three observations that shaped the system: per-dialect adapters outperform a jointly trained adapter on all eight dialects even when the joint model is trained for twice as long; replacing beam search with temperature sampling in order to produce diverse hypotheses for rescoring costs more accuracy than the rescoring recovers; and because the official scorer normalizes hypothesis and reference identically, generic character-level normalization cannot help, whereas learning each dialect's annotation convention from the training transcriptions does.

1 Introduction

NADI 2026 Subtask 1.1 asks for transcription of noisy speech in eight Arabic dialects: Egyptian, Jordanian, Algerian, Emirati, Moroccan, Mauritanian, Yemeni, and Palestinian (Sullivan et al., 2026). Systems are ranked by country-average word error rate. Two properties of the task interact to make it difficult. The recordings are noisy and low-bandwidth, and the training data is small, at roughly 1,600 utterances per dialect. Dialectal Arabic also has no standard orthography (Habash et al., 2012), so the same spoken word may be written in several ways by different annotators.

Multilingual speech models such as Whisper (Radford et al., 2023) and SeamlessM4T (Seamless Communication et al., 2023) cover Arabic, but their coverage of dialects is uneven. Talafha et al. (2023) benchmark Whisper across a range of Arabic varieties and report error rates that stay high for the dialects furthest from Modern Standard Arabic, which matches what we observe for Algerian and Mauritanian. The usual response is more data, whether broadcast speech as in MGB-2 (Ali et al., 2016) or multidialectal speech as in Casablanca (Talafha et al., 2024), and systems built on such collections reach considerably lower error rates than systems restricted to shared-task data.

Our participation is limited to the shared-task data. We use no external speech corpora and no additional pretraining, and our absolute error rates should be read in that light. Within this constraint, our aim was to determine which design choices actually matter and to check each of them against held-out data rather than adopt them by convention.

Orthographic variation has been addressed in this literature mainly by defining a standard to write in. CODA (Habash et al., 2012) proposes a conventional orthography intended for computational work, and later guidelines extend it across dialects (Habash et al., 2018). What has not been done, as far as we are aware, is the inverse: estimating the convention a particular set of annotators actually used and moving system output toward it. That distinction matters here because the scorer applies the same normalization to hypothesis and reference, so a prescribed standard cancels out while the annotators' own convention does not. How to allocate adapter capacity across related varieties is a second open question. Song et al. (2024) attach low-rank modules to Whisper precisely to reduce interference between languages, while also exploiting similarity between languages when extending to new ones, which indicates that interference and useful sharing are both present and that the bal-

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ance between them is empirical rather than settled. Our joint against per-dialect comparison is a small instance of that question for eight dialects of one language, and we report it, including its confound, rather than only the configuration we shipped.

This paper describes the final system and the three observations that determined its shape. The third of these, treating orthographic variation as a question of annotation consistency rather than text normalization, is the one we expect to transfer to other dialectal settings.

2 System

The system combines the `whisper-large-v3-turbo` backbone (Radford et al., 2023), one low-rank adapter per dialect (Hu et al., 2022), beam-search decoding with tuned parameters, and a two-stage text normalization.

2.1 Backbone and adaptation

We use `whisper-large-v3-turbo` (809M parameters) for all eight dialects. Whisper is a reasonable starting point for Arabic dialects but is far from solved on them (Talafha et al., 2023). We also evaluated SeamlessM4T-v2 (Seamless Communication et al., 2023) as an alternative backbone in preliminary experiments and found it weaker on our development data, so we did not pursue it.

We initially expected a single multi-dialect adapter to benefit from sharing information across dialects, since the eight varieties overlap in vocabulary and phonology. The comparison did not support this. Adapters trained independently for each dialect gave lower validation WER on all eight dialects. To rule out the obvious alternative explanation, that the joint adapter was simply undertrained, we retrained it for 20 epochs with the same checkpoint-selection procedure and repeated the comparison. The per-dialect adapters still won on every dialect, as shown in Table 1. We note, however, that the per-dialect configuration also has eight times the total adapter capacity of the joint one. Capacity and interference are therefore confounded in this comparison, and it does not on its own establish negative transfer; we report the result as a comparison between two practical configurations and leave a capacity-matched joint adapter to future work.

Each adapter is initialized from the unmodified backbone rather than continued from a joint checkpoint. Low-rank updates are applied to the query and value projection matrices of every attention block, which leaves 6.55M trainable parameters, or 0.80% of the model. Table 2 lists the adaptation, optimization, and decoding settings.

Dialect	Joint (20 ep.)	Per-dialect
Egyptian	0.6252	**0.5746**
Jordanian	0.6413	**0.6205**
Algerian	0.8707	**0.8467**
Emirati	0.6138	**0.5763**
Moroccan	0.7270	**0.7056**
Mauritanian	0.8194	**0.7986**
Yemeni	0.7855	**0.7543**
Palestinian	0.6811	**0.6339**

Table 1: Validation WER for a jointly trained adapter against per-dialect adapters. The joint adapter was trained for 20 epochs with best-checkpoint selection, so the gap is not explained by under-training. The two configurations differ in total adapter capacity, which is discussed in the text.

Setting	Value
Adapted projections	query, value
Adapter rank	32
Scaling factor α	64
Adapter dropout	0.05
Trainable parameters	6.55M (0.80%)
Optimizer	AdamW
Peak learning rate	10^{-3}
Schedule	cosine, 10% warmup
Batch size	16
Epochs	up to 10, best checkpoint
Decoding	beam search
Beam width	8
Length penalty	-0.5

Table 2: Settings used for every dialect.

We then asked whether the advantage of low-rank adaptation was mainly a regularization effect. Updating all parameters of the backbone offers far more capacity, and it is the approach taken by several systems trained on larger data. On roughly 1,600 utterances per dialect it overfitted sooner than the adapter and gave higher validation WER, 0.5094 against 0.4923 on Jordanian, so we kept low-rank adaptation throughout.

2.2 Decoding

An earlier version of our pipeline used temperature sampling, with temperature 0.7 and nucleus threshold 0.95, in place of beam search. The motivation was rescoring: beam search returned near-identical hypotheses, which left a language model nothing to choose between, and sampling seemed the natural

way to obtain diverse candidates.

Measuring that choice directly showed it to be the wrong trade. Holding everything else fixed, sampling cost 8.2 WER on Egyptian and 8.8 on Moroccan relative to beam search, considerably more than rescoring returned at any point in our experiments. We removed sampling and n-best rescoring from the system.

A second inherited assumption also proved wrong. Two generation constraints, a repetition penalty of 1.5 and a block on repeated trigrams, had been added to suppress loops under greedy decoding and were carried forward without re-examination. Under beam search they suppress legitimate output, since trigram repetition is common in spontaneous dialectal speech. Removing the trigram block alone improved Egyptian WER by 1.8 points. A sweep over the remaining parameters selected a beam width of 8 with a length penalty of -0.5, which we confirmed on a second dialect before adopting.

2.3 Normalization and annotation convention

Our first normalization stage reproduces the preprocessing that the official scorer applies to both sides before error rates are computed. It removes Arabic and Latin punctuation, strips the short-vowel diacritics and the elongation character, deletes the marker that annotators use for unintelligible speech, inserts whitespace between Arabic script and any adjacent Latin letters or digits, replaces underscores and hyphens with spaces, collapses repeated whitespace, and lowercases the remaining Latin characters.

These details matter because word error rate is computed over tokens produced after this step. Where our own preprocessing differs from the scorer's, a correctly recognized word can be split or joined differently from the reference and counted as an error. The segmentation rules were the main source of such mismatches in our case: a hypothesis that writes a numeral immediately followed by an Arabic word disagrees on token boundaries with a reference in which the two are separated, even though the transcription is correct. Matching the scorer removes this class of error at no modeling cost.

We then examined the orthographic variation that is characteristic of dialectal Arabic. Figure 1(b) plots WER against CER for the eight dialects. The ratio between them ranges from 1.7 to 2.5, and for the three most accurate dialects it is close to 2.5, which indicates that a substantial share of scored errors are words differing from the reference by a single character rather than words that were misheard.

The apparent remedy is character-level normalization of alef variants, *ta marbuta*, and *alef maqsura*, in the spirit of conventionalized orthographies for dialectal Arabic (Habash et al., 2012, 2018). This does not help here. The official scorer applies the same normalization to hypothesis and reference, so any gain from collapsing variants on our side alone is removed during evaluation.

We therefore treated the problem as one of annotation consistency rather than text normalization. For a word w, let $S(w)$ be its orthographic skeleton, obtained by collapsing alef variants to bare alef, *alef maqsura* to *ya*, and *ta marbuta* to *ha*. From the training transcriptions of each dialect we estimate $P(v \mid s)$, the distribution over surface forms v that the annotators used for skeleton s, and define the canonical form

$$C(s) = \arg \max_{v:\, S(v)=s} P(v \mid s),$$

keeping the mapping only when s occurs at least $k = 3$ times and the dominant form accounts for at least a fraction $\tau = 0.70$ of those occurrences. At inference, an output word w is rewritten to $C(S(w))$ when a mapping exists and left unchanged otherwise, so unseen vocabulary is never modified. The thresholds were chosen once on validation data and were not tuned per dialect.

This aligns the output with the annotation convention of the dialect instead of applying language-wide spelling rules. Table 3 reports the effect per dialect on the validation set, together with the proportion of utterances the rewriting touched at all. The two dialects where the step is neutral, Algerian and Yemeni, are also the two where almost no mapping survives the thresholds, at 0.0% and 0.7% of utterances, so the absence of an effect is a consequence of orthographic variation within those training transcriptions rather than of the rewriting being harmful. Across the six remaining dialects the size of the gain tracks how often the mapping applies, with the largest change on Jordanian, where 14.7% of utterances are touched. We enabled the step for those six dialects and disabled it for Algerian and Yemeni. The gains are small in absolute terms, which is consistent with the mappings applying to a limited share of utterances.

Dialect	Plain	Rewritten	Touched
Egyptian	0.4864	**0.4834**	5.3%
Jordanian	0.5214	**0.5160**	14.7%
Algerian	0.7883	0.7883	0.0%
Emirati	0.4898	**0.4877**	11.3%
Moroccan	0.6279	**0.6243**	7.3%
Mauritanian	0.7323	**0.7310**	2.7%
Yemeni	0.6886	0.6886	0.7%
Palestinian	0.5043	**0.5020**	5.3%
Average	0.6049	**0.6027**	

Table 3: Validation WER before and after canonicalization, with the proportion of utterances in which at least one word was rewritten. The two dialects with no measurable effect are the two where the thresholds leave almost no mappings.

Dialect	WER	CER
Palestinian	0.4393	0.1744
Emirati	0.4851	0.1931
Jordanian	0.4900	0.1974
Egyptian	0.6095	0.3516
Yemeni	0.6403	0.2853
Moroccan	0.6426	0.2746
Algerian	0.7417	0.3590
Mauritanian	0.7936	0.4028
Country-average	**0.6053**	**0.2798**

Table 4: Final test results per dialect, ordered by WER.

3 Data

We used only the official NADI 2026 Subtask 1.1 release. The train split, approximately 1,600 clips per dialect, was used for adapter training and for building the canonicalization tables. The validation split was used for checkpoint and hyperparameter selection. All audio was resampled to 16 kHz.

4 Results

Figure 1(a) reports final test results per dialect. Country-average WER is 0.6053 and country-average CER is 0.2798, against 0.804 for the zero-shot backbone.

The ordering across dialects was stable throughout our experiments. Palestinian, Emirati, and Jordanian remained the most accurate, Algerian and Mauritanian the hardest by a wide margin, and no configuration we tested changed that ranking. The same two dialects are also the hardest for systems trained on far more data (Talafha et al., 2024), which suggests the difficulty reflects properties of the varieties and their recording conditions rather than our training budget alone.

Egyptian is the clearest exception to this relationship. Its CER is high relative to its WER, with a ratio of 1.7 against 2.5 for Palestinian, which points to errors at the character level rather than to orthographic mismatch. It is also the dialect where canonicalization applied to the fewest utterances, which is consistent with that reading.

5 Discussion

Three things in our results seem worth carrying forward. The first is that the largest single gain we found was not a modeling change at all. Switching the decoder from temperature sampling back to beam search recovered more than eight WER points on two dialects, and removing generation constraints that had been inherited from an earlier greedy configuration recovered close to two more. Both settings had been adopted for reasons that were sound when they were introduced and were never re-examined afterwards. In a low-resource setting where modeling gains are measured in fractions of a point, an unchecked decoding assumption can dominate everything else.

The second is that the orthographic effect is real but bounded. Table 3 shows the gain tracking coverage rather than dialect difficulty: the step helps where the training transcriptions are consistent enough to yield mappings, and does nothing where they are not. This suggests the ceiling for this kind of correction is set by annotation consistency in the training data, and that a dialect with inconsistent references cannot be helped by learning a convention from them. It also suggests where more headroom might lie. Our skeletons collapse only single-character variation, whereas the WER to CER ratios in Figure 1(b) leave room for mismatches at the level of clitic attachment and word segmentation, which the current formulation does not model.

The third is the comparison we cannot yet settle. The per-dialect configuration won on every dialect, but it also carries eight times the adapter capacity of the joint one, and the natural next experiment is a joint adapter whose rank matches the total capacity of the eight per-dialect adapters. If per-dialect adaptation still wins under matched capacity, the result becomes evidence about interference rather than about parameter budget. A related and cheaper experiment is to group dialects by family, for example the Maghrebi varieties against the Levantine ones, and ask whether transfer within a group is positive even where transfer across all eight is not.

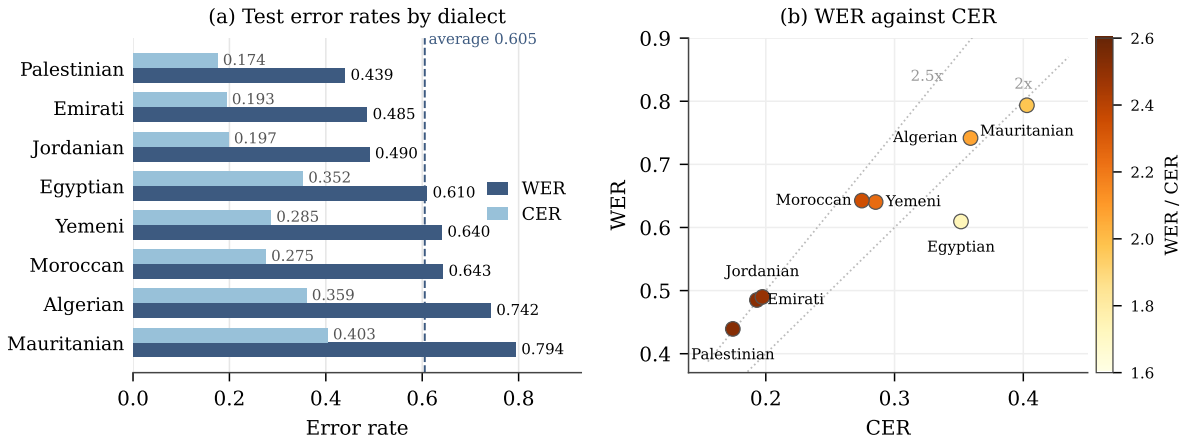


Figure 1: (a) Final test WER and CER per dialect, ordered by WER; the dashed line marks the country-average WER of 0.6053. (b) WER against CER, colored by their ratio, with dotted lines at ratios of 2.0 and 2.5. Dialects near the 2.5 line lose many words to small character differences, which is the pattern the canonicalization step targets. Egyptian is the clear outlier at 1.7.

Both experiments use only shared-task data and would sharpen the claim considerably.

6 Conclusion

We described a system for NADI 2026 Subtask 1.1 built entirely from the shared-task data, combining per-dialect low-rank adaptation of `whisper-large-v3-turbo` with tuned beam-search decoding and a normalization step matched to the annotation conventions of the references. It reaches 0.6053 country-average WER, removing about a quarter of the errors made by the zero-shot backbone, while remaining well behind systems trained with large external corpora.

Two of our observations may be useful beyond this task. Decoding choices made to enable one component can cost more than that component returns, as sampling did in our case. More generally, when an evaluation metric applies the same normalization to hypothesis and reference, it is invariant to generic normalization but still sensitive to the annotation convention, so the useful intervention is to learn that convention from the data rather than to apply general spelling rules. We expect this distinction to matter wherever reference transcriptions are produced by annotators working without a fixed orthographic standard.

Limitations

Our system is trained on approximately 1,600 utterances per dialect and does not use external data. The comparison between full-parameter updating and parameter-efficient adaptation is therefore specific to this training setting. Similarly, the joint and per-dialect adapter configurations differ in their overall adapter capacity, so the comparison reflects two practical design choices rather than parameter sharing alone; separating the two would require a joint adapter with rank matched to the total capacity of the eight per-dialect adapters, and we were not able to run that experiment in the time available. For the same reason we did not test whether particular subsets of dialects transfer positively to one another, which the per-dialect results suggest is worth examining. Checkpoint and configuration selection was based on a small in-domain validation set, and the final results reported here are those obtained on the test set. Finally, the canonicalization step relies on sufficient consistency within the training transcriptions for a dominant spelling pattern to emerge. This assumption was supported for six of the eight dialects, while the remaining two showed greater orthographic variation.

Code Availability

The code for training the adapters, tuning the decoding settings, building the canonicalization tables, and producing the final submission is available at https://github.com/mohamedkamalsabaa/Sabaa-at-NADI-2026-Subtask-1.1.

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